Topic Experiment - 7 Date Page No

Aim: Analyse a series RLC circuit and draw the phasor diagrams.

Apparatus: Digital multimeter, variable resistor, inductor, capacitor, AC power supply, connecting wires.

Theory: Impedance (Z), $Z = \sqrt{R^2 + (X_L - X_C)^2}$

V_L (Voltage drop across inductor) = $I X_L$

V_C (Voltage drop across capacitor) = $I X_C$

V_L leads I by $\pi/2$, and V_C lags I by $\pi/2$

If $X_C > X_L$; then the circuit is capacitive

If $X_L > X_C$; then the circuit is inductive

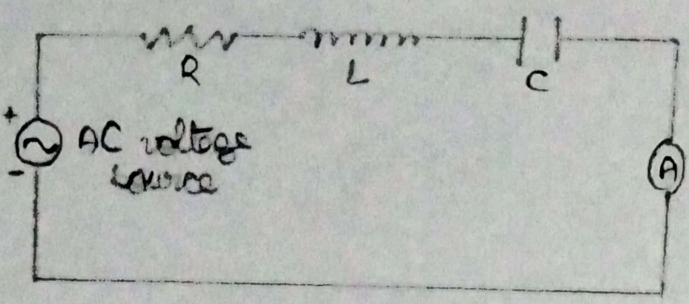
$$\tan\phi = \frac{|V_L - V_C|}{V_R} = \frac{|X_L - X_C|}{R}$$

$$\cos\phi \ (P.f) = \frac{R}{Z} \ ; \quad \boxed{P = V_{rms} I_{rms} \cos\phi}$$

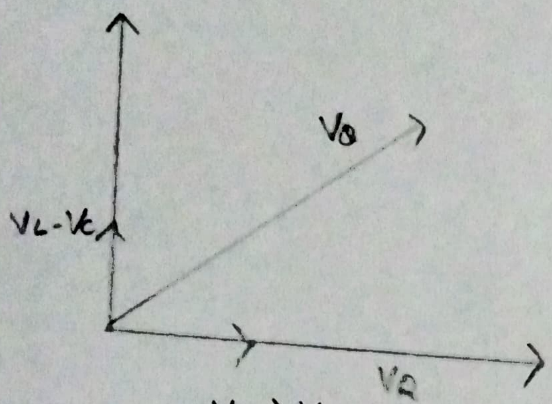
Procedure:
1) Connect the circuit as shown in the circuit diagram
2) Measure AC voltage V from the mains using digital multimeter.
3) Measure the circuit current I using another multimeter connected in series in the circuit.
4) Note the voltage drop V_R, V_L and V_C using the digital multimeter.
5) Using above parameters find the value of R, X_L, L, X_C, C, Z and $\cos\phi$

Result: Impedance (Z) of RLC circuit is 55 Ω and power factor is 0.77 (lagging)

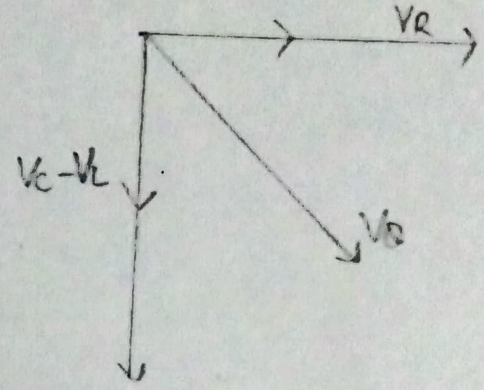
Circuit Diagram



Phasor Diagram



$V_L > V_C$
$X_L > X_C$
$\theta > 0°$



$V_L < V_C$
$X_L < X_C$
$\theta < 0°$

Observations and Calculations

Observed value

V (V)	V_R (V)	V_L (V)	V_C (V)	I (A)
220V	170V	150V	100V	4A

Calculated value

R (Ω)	X_L (Ω)	X_C (Ω)	Z (Ω)	cos ϕ	P
42.5	37.5	25	55	0.77	677.6

Expected Outcome Attained: The series RLC circuit was analysed and the corresponding phasor diagram was drawn.